Week 5 Discussion

Covers: the probability integral transformation and expected values (Lecture 4.2); moments, variance, and mgfs (Lecture 5.1); what the mgf determines.

Problem 1 (The probability integral transformation)

Let X have pdf

$$f_X(x) = \begin{cases} \dfrac{x-1}{2} & 1 < x < 3 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the cdf $F_X(x)$. Then find a monotone function $u(x)$ such that the random variable $Y = u(X)$ has a uniform(0, 1) distribution. Verify your answer directly, by computing $P(Y \le y)$ for $0 < y < 1$. *Hint:* Theorem 2.1.10.

(b) Now go the other way. Let $V \sim$ uniform(0, 1). Find a function h such that $h(V)$ has the same distribution as X, and check that $P(h(V) \le x) = F_X(x)$ for $1 < x < 3$. How would you use this to simulate X on a computer?

(c) Find $\mathrm{E}X$ in two ways:

- directly from f_X, by (2.2.4);
- as $\mathrm{E}\, h(V)$, using your answer to (b) and the uniform pdf.

Then use the fact that $\mathrm{E}F_X(X) = \frac{1}{2}$ to find $\mathrm{E}(X-1)^2$ **without** integrating against f_X, and deduce $\mathrm{Var}\, X$.

(d) Theorem 2.1.10 needs a *continuous* cdf. Let X be uniform on $\{1, 2, 3\}$, that is, $P(X = k) = \frac{1}{3}$ for $k = 1, 2, 3$, and let $Y = F_X(X)$.

- Find the pmf of Y and $\mathrm{E}Y$. Is Y uniform(0, 1)?
- Show that $F_Y(y) \le y$ for all $0 < y < 1$, with strict inequality for some y. (C&B say Y is *stochastically greater* than a uniform(0, 1).)
- Where does the proof of Theorem 2.1.10 use continuity of F_X, and why does that step fail here?

> **(a)** For $1 < x < 3$,
>
> $$F_X(x) = \int_1^x \frac{s-1}{2}\, ds = \frac{(x-1)^2}{4},$$

with $F_X(x) = 0$ for $x \le 1$ and $F_X(x) = 1$ for $x \ge 3$.

- By Theorem 2.1.10, take $u(x) = F_X(x) = \frac{(x-1)^2}{4}$. It is increasing on $(1, 3)$.
- *Direct check*. For $0 < y < 1$, and since $X > 1$,

$$\begin{aligned} P(Y \le y) &= P\left(\frac{(X-1)^2}{4} \le y\right) = P\big(X \le 1 + 2\sqrt{y}\big) \\ &= F_X\big(1 + 2\sqrt{y}\big) = \frac{(2\sqrt{y})^2}{4} = y. \end{aligned}$$

(b) Invert F_X: solving $v = \frac{(x-1)^2}{4}$ for $x \in (1, 3)$ gives $F_X^{-1}(v) = 1 + 2\sqrt{v}$. So take $h(v) = 1 + 2\sqrt{v}$. For $1 < x < 3$,

$$P\big(1 + 2\sqrt{V} \le x\big) = P\left(V \le \frac{(x-1)^2}{4}\right) = \frac{(x-1)^2}{4} = F_X(x).$$

To simulate X: generate V from uniform(0, 1) and return $1 + 2\sqrt{V}$. This is the inverse cdf method from Lecture 4.2.

(c)

- **Directly:** substituting $s = x - 1$,

$$\mathrm{E}X = \int_1^3 x \cdot \frac{x-1}{2}\, dx = \frac{1}{2}\int_0^2 (s+1)s\, ds = \frac{1}{2}\left(\frac{8}{3} + 2\right) = \frac{7}{3}.$$

- **Through V:** X has the same distribution as $1 + 2\sqrt{V}$, so

$$\mathrm{E}X = \mathrm{E}\big(1 + 2\sqrt{V}\big) = 1 + 2\int_0^1 v^{1/2}\, dv = 1 + 2 \cdot \frac{2}{3} = \frac{7}{3}.$$

- **Second moment for free:** $F_X(X) = \frac{(X-1)^2}{4}$, and $F_X(X) \sim$ uniform(0, 1) by (a). So

$$\mathrm{E}(X-1)^2 = 4\,\mathrm{E}F_X(X) = 4 \cdot \frac{1}{2} = 2.$$

- **Variance:** by Theorem 2.3.4 (a shift does not change the variance) and (2.3.1),

$$\mathrm{Var}\, X = \mathrm{Var}(X-1) = \mathrm{E}(X-1)^2 - \big(\mathrm{E}(X-1)\big)^2 = 2 - \left(\frac{4}{3}\right)^2 = \frac{2}{9}.$$

Check: $\int_1^3 (x-1)^2 \frac{x-1}{2}\, dx = \frac{1}{2} \cdot \frac{2^4}{4} = 2$. ✓

(d)

- $F_X(1) = \frac{1}{3}$, $F_X(2) = \frac{2}{3}$, and $F_X(3) = 1$. So Y takes the values $\frac{1}{3}$, $\frac{2}{3}$, 1, each with probability $\frac{1}{3}$, and

$$\mathrm{E}Y = \frac{1}{3}\left(\frac{1}{3} + \frac{2}{3} + 1\right) = \frac{2}{3} \neq \frac{1}{2}.$$

  Y is discrete, so it is certainly not uniform(0, 1).

- For $0 < y < 1$, $F_Y(y) = \frac{\lfloor 3y \rfloor}{3} \leq y$. The inequality is strict except at $y = \frac{1}{3}$ and $y = \frac{2}{3}$. For example, $F_Y\left(\frac{1}{2}\right) = \frac{1}{3} < \frac{1}{2}$.

- The last step of the proof, $F_X\left(F_X^{-1}(y)\right) = y$, is labeled "continuity of F_X." It needs every level $y \in (0, 1)$ to be hit by F_X. Here F_X jumps from $\frac{1}{3}$ to $\frac{2}{3}$ at $x = 2$, so no x has $F_X(x) = \frac{1}{2}$. By (2.1.13), $F_X^{-1}\left(\frac{1}{2}\right) = 2$ and $F_X(2) = \frac{2}{3} \neq \frac{1}{2}$. The jumps push $F_X(X)$ upward, which is why Y is stochastically greater than a uniform.

Problem 2 (MGF)

Let X have the geometric pmf

$$P(X = x) = p(1-p)^x, \qquad x = 0, 1, 2, \ldots, \quad 0 < p < 1.$$

X counts the failures before the first success in independent trials with success probability p. (Week 4 Discussion Problem 2 used $p = \frac{1}{3}$.)

(a) Verify that

$$M_X(t) = \frac{p}{1 - (1-p)e^t}.$$

For which t does the sum converge? Does the mgf exist in the sense of Definition 2.3.6?

(b) Use Theorem 2.3.7 to find $\mathrm{E}X$, $\mathrm{E}X^2$, and $\mathrm{Var}\, X$.

(c) (C&B Exercise 2.32) Let $S(t) = \log M_X(t)$, for any random variable X with an mgf. Show that

$$S'(0) = \mathrm{E}X \quad \text{and} \quad S''(0) = \mathrm{Var}\, X.$$

Then redo (b) for the geometric with $S(t)$. Which way is less work?

(a) Write $q = 1 - p$. Then

$$M_X(t) = \sum_{x=0}^{\infty} e^{tx} p q^x = p \sum_{x=0}^{\infty} \left(qe^t\right)^x = \frac{p}{1 - qe^t},$$

provided $qe^t < 1$, that is, $t < -\log(1-p)$.

- Since $0 < q < 1$, the bound $-\log(1-p)$ is **positive**. So the sum is finite on $(-h, h)$ with $h = -\log(1-p)$, and the mgf exists.
- For $t \geq -\log(1-p)$ the terms do not go to 0, and the sum is infinite. Like the gamma mgf (Example 2.3.8), it is finite only on one side of a cutoff, but the cutoff is away from 0.

(b) Let $D(t) = 1 - qe^t$, so $D(0) = p$. Then

$$M_X'(t) = \frac{pqe^t}{D(t)^2},$$
$$M_X''(t) = \frac{pqe^t}{D(t)^2} + \frac{2pq^2e^{2t}}{D(t)^3}.$$

At $t = 0$:

$$\mathrm{E}X = M_X'(0) = \frac{pq}{p^2} = \frac{q}{p},$$
$$\mathrm{E}X^2 = M_X''(0) = \frac{q}{p} + \frac{2q^2}{p^2} = \frac{q(2-p)}{p^2}.$$

By (2.3.1),

$$\operatorname{Var} X = \frac{q(2-p)}{p^2} - \frac{q^2}{p^2} = \frac{q(2-p-q)}{p^2} = \frac{1-p}{p^2}.$$

(c) By the chain rule, $S'(t) = \dfrac{M_X'(t)}{M_X(t)}$, and by the quotient rule

$$S''(t) = \frac{M_X''(t)M_X(t) - \big(M_X'(t)\big)^2}{M_X(t)^2}.$$

Since $M_X(0) = 1$, $M_X'(0) = \mathrm{E}X$, and $M_X''(0) = \mathrm{E}X^2$,

$$S'(0) = \mathrm{E}X, \qquad S''(0) = \mathrm{E}X^2 - (\mathrm{E}X)^2 = \operatorname{Var} X.$$

For the geometric, $S(t) = \log p - \log(1 - qe^t)$, so

$$S'(t) = \frac{qe^t}{1-qe^t}, \qquad S''(t) = \frac{qe^t}{(1-qe^t)^2}.$$

- At $t = 0$: $S'(0) = \frac{q}{p}$ and $S''(0) = \frac{q}{p^2}$, as in (b).
- S is less work. The logarithm turns the constant p into an additive term that disappears when differentiated, and it gives $\operatorname{Var} X$ directly, with no subtraction step.

Problem 3 (MGF)

(a) Each function below is claimed to be the mgf of some random variable X. Identify the distribution of X, or explain why no such X exists. *Hint:* Theorems 2.3.11 and 2.3.15, and Examples 2.3.8 and 2.3.9.

- **(i)** $M(t) = \left(\frac{1}{3} + \frac{2}{3}e^t\right)^5$, for all t.
- **(ii)** $M(t) = \dfrac{e^{2t}}{1-3t}$, for $t < \frac{1}{3}$.
- **(iii)** $M(t) = \dfrac{t}{1-t}$, for $|t| < 1$.

(b) For $n = 2, 3, \ldots$, let X_n have the geometric pmf of Problem 2 with $p = \frac{1}{n}$, and let $Y_n = X_n/n$. Think of trials run every $\frac{1}{n}$ seconds, each succeeding with probability $\frac{1}{n}$, so that Y_n is the waiting time, in seconds, before the first success.

- Use Problem 2(a) and Theorem 2.3.15 to find $M_{Y_n}(t)$.
- Show that $M_{Y_n}(t) \to \frac{1}{1-t}$ for every $t < 1$, and say what Theorem 2.3.12 lets you conclude about F_{Y_n}. *Hint:* $n\left(e^{t/n} - 1\right) \to t$.

(a) By Theorem 2.3.11(b), once we find one distribution with a given mgf, it is the only one.

- **(i)** This is Example 2.3.9 with $n = 5$ and $p = \frac{2}{3}$, so $X \sim$ binomial(5, $\frac{2}{3}$).
- **(ii)** $\frac{1}{1-3t}$, for $t < \frac{1}{3}$, is the exponential(3) mgf (Example 2.3.8 with $\alpha = 1$, $\beta = 3$). By Theorem 2.3.15 with $a = 1$ and $b = 2$, $e^{2t}M_Z(t)$ is the mgf of $Z + 2$. So X has the same distribution as $Z + 2$, with $Z \sim$ exponential(3), and

$$f_X(x) = \frac{1}{3}e^{-(x-2)/3}, \qquad x > 2.$$

- **(iii)** No such X exists. Every mgf has $M_X(0) = \mathrm{E}e^0 = 1$, but here $M(0) = 0$.

(b)

- With $a = \frac{1}{n}$ and $b = 0$ in Theorem 2.3.15, $M_{Y_n}(t) = M_{X_n}(t/n)$, so

$$M_{Y_n}(t) = \frac{1/n}{1 - \left(1 - \frac{1}{n}\right)e^{t/n}} = \frac{1}{n\left[1 - \left(1 - \frac{1}{n}\right)e^{t/n}\right]}.$$

  By Problem 2(a), this is finite for $t < -n\log(1 - 1/n)$, and $-n\log(1 - 1/n) > 1$. So $M_{Y_n}(t)$ is defined for every $t < 1$.

- Expanding the denominator,

$$n\left[1-\left(1-\frac{1}{n}\right)e^{t/n}\right] = -n\big(e^{t/n}-1\big)+e^{t/n}$$
$$\to -t+1.$$

Hence $M_{Y_n}(t)\to \frac{1}{1-t}$ for every $t<1$, which includes the neighborhood $|t|<1$ of 0.

- $\frac{1}{1-t}$ is the mgf of the exponential(1) distribution, with cdf $F(y)=1-e^{-y}$, $y\geq 0$. Since this F is continuous, Theorem 2.3.12 gives $F_{Y_n}(y)\to 1-e^{-y}$ for **every** y. Waiting for a rare success in many quick trials takes an approximately exponential time.